

SSA1 Plasma Membrane (GO:0005886)

Annotation — Deep Research Report

Gene: SSA1 (P10591) · *Saccharomyces cerevisiae* (NCBITaxon:559292) **Focus:** existing_go_annotation_decision · existing_annotations[3] **Hypothesis slug:** existing-go-0005886-keep-as-non-core **Term under review:** plasma membrane (GO:0005886), evidence IBA, original reference GO_REF:0000033 **Seed action:** KEEP_AS_NON_CORE

Summary

Executive verdict: partially supported — the recommended action (KEEP_AS_NON_CORE) is defensible on substance, but the rationale stated in the seed is factually wrong and must be corrected before the review is finalized. The seed hypothesis asserts that (a) the plasma-membrane annotation is an IBA (phylogenetically inferred) term that is "consistent" with an underlying HDA detection, and (b) it should be retained conservatively as non-core. A live evidence check contradicts point (a): the PANTHER/PAINT phylogenetic (IBA) pipeline for P10591 propagated only nucleus (GO:0005634), cytoplasm (GO:0005737), and cytosol (GO:0005829) — **it never assigned plasma membrane**. The GO:0005886 annotation rests entirely on a single high-throughput proteomics dataset (HDA, ECO:0007005, [PMID: 16622836](#)), a crude "stripped plasma membrane" fraction in which Ssa1 was one of ~90 co-purifying proteins and was not functionally highlighted. The annotation is therefore even weaker than the seed's own summary implies.

Ssa1 is, by every line of primary evidence, an **abundant cytosolic Hsp70 chaperone**. Its sequence (UniProt P10591, 642 aa) has no transmembrane segment, no signal peptide, no GPI anchor, and no lipidation site, and its UniProt keyword set contains Cytoplasm, Cell wall, and Secreted but **not** Membrane/Cell membrane. Where Ssa1 genuinely contacts membranes, it does so as a peripheral, cytosol-facing chaperone — recruited to the vacuolar membrane during Cvt/aminopeptidase transport ([PMID: 10745074](#)), acting at the cytosolic face of the ER during ERAD ([PMID: 18812321](#)), and delivering precursors to the mitochondrial import receptors ([PMID: 27412066](#)). None of these establish integral or stable plasma-membrane residence.

Bottom line for the curator: the plasma-membrane term should not be treated as core, so KEEP_AS_NON_CORE is an acceptable conservative outcome — but the review's supporting text should be rewritten to (1) remove the incorrect claim that an IBA supports plasma membrane, (2) state accurately that the term rests on one crude proteomic fraction (HDA, not IBA), and (3) note that REMOVE is a fully defensible alternative given the absence of any functional, genetic, or targeted-localization support. If a genuine surface pool of Ssa1 exists, the better-supported descriptor is the fungal-type cell wall (GO:0009277, IDA), not the plasma membrane.

Key Findings

Finding 1 — The plasma-membrane term rests only on high-throughput fraction proteomics; KEEP_AS_NON_CORE is justified on the substance

The plasma-membrane cellular-component assignment for Ssa1 traces to a single source: *The plasma membrane proteome of Saccharomyces cerevisiae and its response to the antifungal calcofluor* (PMID: 16622836, Delom et al., 2006). This study prepared a "**stripped plasma membrane fraction**" — a crude enrichment that is solubilized with detergent and then subjected to mass spectrometry. The verified snippet from the abstract reads: "*Proteins from a stripped plasma membrane fraction were solubilized with the neutral and non-denaturing detergent.*" Ssa1 was one of roughly 90 identified proteins and was not singled out for any membrane-specific function.

This kind of subcellular-fraction proteomics is precisely the situation in which an abundant, sticky cytosolic chaperone will appear as a contaminant or peripheral associate. The primary literature is unanimous that Ssa1's core identity is cytosolic. A study of ERAD of the thiazide-sensitive NaCl cotransporter notes the process "*required the activity of Ssa1, a cytoplasmic Hsp70 chaperone*" (PMID: 22027832). Work on tombusvirus replication describes Hsp70 as "*an abundant cytosolic chaperone*" (PMID: 19153242). High cellular abundance combined with intrinsic client-binding stickiness make co-fractionation with any membrane preparation a strong prior expectation. Because the annotation reflects bulk co-purification rather than a demonstrated functional site, treating plasma membrane as **non-core is the correct handling of the evidence weight** — the seed's action is substantively defensible.

Finding 2 — There is NO IBA plasma-membrane annotation; the seed's "IBA is consistent" rationale is incorrect

A live QuickGO enumeration of all cellular-component annotations for P10591 returns 16 CC annotations. Critically, the phylogenetic IBA pipeline (GO_REF:0000033, contributed by GO_Central via PANTHER/PAINT) assigned only three locations: **GO:0005634 nucleus**, **GO:0005737 cytoplasm**, and **GO:0005829 cytosol**. It did **not** assign GO:0005886 plasma membrane. The only plasma-membrane annotation present is a single **HDA** (ECO:0007005), qualifier `located_in`, referencing

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assigned by SGD.

This directly contradicts the seed context, which lists the term as `evidence_type: IBA` with the summary "IBA is consistent." There is no IBA to be consistent with. The seed appears to have conflated the GO_REF:0000033 phylogenetic reference (which is attached to the *nucleus/cytoplasm/cytosol* propagations) with the plasma-membrane term. The plasma-membrane annotation is neither phylogenetically supported nor independently replicated — it is one crude proteomics hit.

By contrast, the core locations for Ssa1 carry **convergent, multi-method support**:

Location	GO ID	Evidence chain
Cytosol	GO:0005829	IBA + HDA (P 26928762)
Cytoplasm	GO:0005737	IBA + HDA (P 11914276 + IDA (P 8755907)
Nucleus	GO:0005634	IBA + IDA (P 10347213 + HDA (P 11914276)
Plasma membrane	GO:0005886	HDA only (P 16622836)

Sequence analysis reinforces the point. UniProt P10591 (642 aa) shows **no transmembrane helix, no intramembrane region, no signal peptide, no GPI-anchor site, and no lipidation modification**. Its keyword set lists Cytoplasm, Cell wall, and Secreted — but **not** Membrane or Cell membrane. A protein that stably resides in the plasma membrane would ordinarily carry at least one such feature or keyword; Ssa1 carries none. This is fully consistent with peripheral/contaminant detection rather than genuine membrane residence.

Mechanistic Model / Interpretation

Ssa1 is a constitutively expressed member of the yeast cytosolic **Ssa (Hsp70) subfamily**. Its molecular function is ATP-dependent chaperone activity: binding exposed hydrophobic segments of client polypeptides to assist folding, prevent aggregation, drive translocation across organellar membranes, and target misfolded proteins for degradation. All of these activities occur in the **cytosol** or **at the cytosolic face of membranes**, not within the membrane bilayer.

SSA1 (cytosolic Hsp70) – where it really acts

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| CYTOSOL (core; IBA+HDA+IDA) |
| - folding / anti-aggregation of nascent & stress-denatured |
| clients |
| - NUCLEUS pool (core; IBA+IDA+HDA) - e.g. Pdr3 co-purification |
| and transcriptional modulation (<a href="https://pubmed.ncbi.nlm.nih.gov/17636264/" |
| |
| -- peripheral, cytosol-FACING membrane contacts (transient) -- |
| - ER surface -> ERAD of CytoQC/membrane clients (<a href="https://pubmed.ncbi.nlm.nih.gov/17636264/" |
| - Vacuole membrane -> Ape1/Cvt vesicle fusion (<a href="https://pubmed.ncbi.nlm.nih.gov/17636264/" |
| - Mito TOM receptors -> precursor delivery (<a href="https://pubmed.ncbi.nlm.nih.gov/27636264/" |
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      ^
      | co-fractionation / peripheral stickiness
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| PLASMA MEMBRANE (GO:0005886) - HDA only, <a href="https://pubmed.ncbi.nlm.nih.gov/1662283/" |
| crude "stripped PM" proteome; Ssa1 = 1 of ~90 proteins; |
| no TM / signal / GPI / lipidation; NOT phylogenetically propagated |
| -> reflects abundance-driven co-purification, not residence |
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The key mechanistic distinction: **a cellular-component annotation is meant to record where the gene product carries out its function.** For Ssa1, that answer is overwhelmingly the cytosol (with nuclear and peripheral membrane pools). The plasma-membrane detection is a *downstream observation of a fractionation experiment*, not evidence of a functional plasma-membrane pool. Even the organism-relevant surface literature points elsewhere: Ssa1 carries the UniProt keywords Cell wall and Secreted, and SGD records a fungal-type cell wall IDA (GO:0009277). Cytosolic Hsp70s are known to reach the yeast surface/cell wall under some conditions, so if any genuine surface pool exists, **fungal-type cell wall (GO:0009277)** is the more informative and better-evidenced descriptor than plasma membrane.

Evidence Matrix

Citation	Evidence type	Supports / Refutes / Qualifies	Claim tested	Key finding	Context	Confidence & limitations
PMID: 16622836	Localization (HT proteomics)	Qualifies (sole basis for the term)	Is Ssa1 at the plasma membrane?	Ssa1 detected in a crude "stripped plasma membrane" fraction; 1 of ~90 proteins, not functionally highlighted	<i>S. cerevisiae</i> , PM fraction MS	Low specificity; a single protein; abundant cytosolic contaminants expected in fractions
QuickGO/ UniProt P10591 (database)	Computational/ database	Refutes seed rationale	Does an IBA support PM?	IBA (GO_REF:0000033) propagated only nucleus, cytoplasm, cytosol — never plasma membrane	GO_Central PANTHER/ PAINT	Direct enumeration of database carry-over; the mechanism to watch
UniProt P10591 features (database)	Structural/ sequence	Refutes membrane residence	Does sequence support PM residence?	No TM / signal / GPI / lipidation; keywords lack Membrane/Cell membrane (has Cytoplasm, Cell wall, Secreted)	642-aa Hsp70	Strong negative prior; absence of feature is absolute proof
PMID: 22027832	Mutant phenotype / functional	Refutes PM as primary site	Where does Ssa1 function?	ERAD of NCC "required the activity of Ssa1, a cytoplasmic Hsp70 chaperone"	<i>S. cerevisiae</i> ERAD	Defines core identity of cytoplasmic
PMID: 19153242	Functional / localization	Refutes PM as primary site	Core compartment of Ssa1/Hsp70	"an abundant cytosolic chaperone"; recruited to	Tombusvirus in yeast	Abundant explains fractionation

Citation	Evidence type	Supports / Refutes / Qualifies	Claim tested	Key finding	Context	Confidence & limitations
				intracellular membranes for viral replication		
PMID: 18812321	Mutant phenotype / functional	Qualifies (peripheral membrane role)	Membrane engagement mode	Ssa1p acts at the cytosolic face of the ER in ERAD of a cytosolic substrate	<i>S. cerevisiae</i> CytoQC	Peripheral not integ not PM
PMID: 10745074	Localization / functional	Qualifies / competing (other membrane)	Where does Ssa1 bind membranes?	Ssa1/2p "prominently localized to the vacuolar membrane" for Ape1 vesicle fusion	<i>S. cerevisiae</i> Cvt pathway	Vacuolar plasma, membra
PMID: 27412066	Interaction / crosslinking	Qualifies (peripheral role)	Ssa1 at organellar surfaces	Ssa1 delivers precursors to mitochondrial import receptors Tom20/Tom70	<i>S. cerevisiae</i> mito import	Cytosolic factor at surface; PM
PMID: 17636264	Interaction / functional	Qualifies (nuclear/ cytosolic role)	Additional Ssa1 compartments	Cytosolic Hsp70 Ssa1p represses PDR5 via nuclear Pdr3	<i>S. cerevisiae</i> Pdr pathway	Supports cytosolic nuclear, PM

GO Curation Implications

Term: plasma membrane (GO:0005886), a cellular-component (CC) term. This is a location claim, not an MF or BP claim.

Lead action (requires curator verification): KEEP_AS_NON_CORE is acceptable, but the review text must be corrected; REMOVE is a defensible alternative.

1. **Do not treat GO:0005886 as core.** No functional, genetic, or targeted-imaging evidence places Ssa1's activity at the plasma membrane. The only support is a single crude-fraction proteomics dataset. Non-core handling is correct in spirit.
2. **Correct the rationale.** The seed's summary ("detected... by HDA; IBA is consistent") is inaccurate. There is **no IBA** to plasma membrane; the PAINT pipeline propagated only nucleus/cytoplasm/cytosol. The evidence chain is HDA-only. The rewritten reason should say: *"Supported only by one high-throughput stripped-plasma-membrane proteomic fraction (*

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); no phylogenetic (IBA), genetic, or targeted-localization support; sequence has no membrane-targeting features. Retained conservatively as non-core; REMOVE is defensible."
3. **Consider REMOVE as the stronger scientific action.** Given (i) HDA-only support, (ii) the high false-positive rate for abundant cytosolic chaperones in membrane fractions, and (iii) zero membrane-targeting sequence features, a curator could reasonably remove the annotation as a fractionation artifact / database carry-over rather than retain it. This should be flagged as an explicit option.
4. **Prefer a better-supported surface descriptor if any surface pool is to be captured.** Fungal-type cell wall (GO:0009277) carries IDA support and matches the UniProt Cell wall / Secreted keywords, making it a more informative CC term than plasma membrane for any genuine surface localization.

The well-supported core CC terms to retain as-is: cytosol (GO:0005829), cytoplasm (GO:0005737), and nucleus (GO:0005634).

Mechanistic Scope

A CC annotation records *the site at which the gene product performs its activity*. Ssa1's activity is ATP-dependent Hsp70 chaperoning, which occurs in the cytosol and nucleus and, transiently, at the cytosolic faces of the ER, vacuolar, and mitochondrial membranes. The plasma-membrane term does not describe any Ssa1 activity; it is an inference drawn from a bulk fractionation experiment. Separating direct activity from observation:

- **Direct gene-product activity/location:** cytosolic (and nuclear) Hsp70 chaperoning — folding, translocation, degradation targeting.
 - **Peripheral (cytosol-facing) membrane engagement:** ER surface (ERAD), vacuolar membrane (Cvt), mitochondrial import receptors — real but not integral, and not plasma membrane.
 - **Downstream/observational only:** presence in a stripped-PM proteomic fraction — an artifact-prone readout, not a functional assignment.
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Conflicts and Alternatives

- **Seed vs. live database (the central conflict):** The seed labels the term IBA and calls the IBA "consistent" with the HDA; the live QuickGO record shows no IBA plasma-membrane annotation at all. This is a database carry-over / evidence-code mislabeling that the curator must reconcile.
 - **Abundance/co-fractionation artifact:** Ssa1 is one of the most abundant cytosolic proteins in yeast. Fraction-based proteomics reliably co-enriches such proteins. This is the leading alternative explanation for the PM detection and is strongly favored over genuine residence.
 - **Paralog considerations:** Ssa1 shares near-identity with Ssa2 (>95%) and functional overlap with Ssa3/Ssa4. Proteomic peptide assignments among Ssa paralogs can be ambiguous, further weakening a single-protein PM claim. The vacuolar-membrane study explicitly treats Ssa1/2p together ([PMID: 10745074](#)).
 - **Alternative surface term:** If any true cell-surface pool exists, fungal-type cell wall (GO:0009277, IDA) is better supported than plasma membrane and is consistent with the Secreted/Cell wall keywords.
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Limitations and Knowledge Gaps

- **No targeted imaging evidence was retrievable** for Ssa1 at the plasma membrane (e.g., GFP tagging with a PM marker, immuno-EM, or fractionation with quantitative controls). What was checked: SGD/UniProt/QuickGO annotation records and primary functional literature. Why it matters: a curator deciding KEEP vs REMOVE would benefit from any high-resolution localization; none supports PM. Resolution: a targeted GFP-Ssa1 co-localization with a PM marker (e.g., Pma1) under native abundance.
 - **The Delom et al. dataset was not re-analyzed at the peptide level.** What was checked: the abstract and its role as the HDA source. Why it matters: knowing whether the Ssa1 identification is peptide-robust and paralog-unambiguous affects confidence. Resolution: inspect the paper's supplementary protein list and spectral counts.
 - **Quantitative contamination controls unknown.** High-throughput PM fractions rarely report cytosolic-contaminant enrichment ratios for individual proteins. Resolution: compare Ssa1's PM-fraction abundance to its total-cell abundance; a chaperone present at cytosolic levels in a PM fraction indicates contamination.
 - **GO_REF:0000033 provenance mismatch.** The seed attaches this phylogenetic reference to the PM term, but the live pipeline attaches it to nucleus/cytoplasm/cytosol. Resolution: curator to verify the current GAF/annotation source and correct the evidence code.
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Proposed Follow-up Experiments / Actions (Discriminating Tests)

1. **Quantitative co-fractionation control:** Measure Ssa1 abundance in the stripped-PM fraction relative to a bona fide integral PM protein (e.g., Pma1) and to total lysate. If Ssa1 tracks cytosolic markers, the PM annotation is an artifact → supports REMOVE.
2. **High-resolution live imaging:** GFP-Ssa1 expressed at native levels co-imaged with a PM marker (Pma1-mCherry). Absence of PM rim enrichment argues against residence.
3. **Carbonate/salt extraction:** Peripheral/contaminant proteins are stripped from membranes by Na_2CO_3 or high salt; integral proteins are not. Ssa1 should extract fully.
4. **Paralog-resolved MS:** Re-examine the Delom peptides for Ssa1-unique vs Ssa-shared peptides to test whether the identification is even Ssa1-specific.

5. **Cell-wall vs PM discrimination:** Spheroplasting and cell-wall digestion controls (or surface biotinylation of intact spheroplasts) to test whether the surface signal is cell-wall-associated (GO:0009277) rather than PM.

Curation Leads (require curator verification)

- **Action:** Retain KEEP_AS_NON_CORE **or** change to REMOVE — both defensible. Prefer REMOVE if the curator weights artifact risk heavily; prefer KEEP_AS_NON_CORE if conservative retention is house policy. Either way, **rewrite the rationale**.
- **Rationale replacement (candidate text):** "GO:0005886 is supported only by one high-throughput 'stripped plasma membrane' proteomic fraction (P 16622836) in which Ssa1 was one of ~90 co-purifying proteins. No phylogenetic (IBA), genetic, or targeted-localization evidence supports plasma-membrane residence; the sequence lacks any transmembrane, signal, GPI, or lipidation feature. Ssa1 is an abundant cytosolic Hsp70; the detection most likely reflects co-fractionation. Retained as non-core (or removable as artifact)."
- **Correct the evidence code:** The term should be recorded as **HDA (ECO:0007005)**, **not IBA**. Verify against the current QuickGO/GAF record for P10591.
- **Candidate references with exact snippets to verify:**
 - **PMID: 16622836:** "Proteins from a stripped plasma membrane fraction were solubilized with the neutral and non-denaturing detergent" — confirms the crude-fraction basis.
 - **PMID: 22027832:** "required the activity of Ssa1, a cytoplasmic Hsp70 chaperone" — confirms cytosolic core identity.
 - **PMID: 19153242:** "an abundant cytosolic chaperone" — explains co-fractionation.
- **Candidate alternative CC term:** fungal-type cell wall (GO:0009277, IDA) if a surface pool is to be represented — better supported than plasma membrane.
- **Suggested curator questions:** Is there any single-gene, low-throughput imaging/genetic evidence for Ssa1 at the PM? Is the Delom identification paralog-specific and abundance-controlled? Does house policy retain single-HDA fraction hits as non-core, or remove them?
- **Suggested experiments:** quantitative PM-fraction contamination control; GFP-Ssa1 native imaging vs Pma1; carbonate extraction; paralog-resolved MS (as in Discriminating Tests).

Evidence Base (key literature)

- [PMID: 16622836](#) — *The plasma membrane proteome of Saccharomyces cerevisiae and its response to the antifungal calcofluor*. The sole source of the GO:0005886 annotation; a crude stripped-PM proteomic fraction. Supports the annotation only weakly (bulk detection).
- [PMID: 22027832](#) — Defines Ssa1 as "a cytoplasmic Hsp70 chaperone" in ERAD. Establishes cytosolic core identity.
- [PMID: 19153242](#) — "an abundant cytosolic chaperone" recruited to intracellular (not plasma) membranes; abundance explains fraction contamination.
- [PMID: 18812321](#) — Ssa1p functions at the cytosolic face of the ER in CytoQC/ERAD; peripheral, not integral, membrane engagement.
- [PMID: 10745074](#) — Ssa1/2p localize to the vacuolar membrane for Ape1 vesicle fusion; genuine membrane contact, but vacuolar not plasma.
- [PMID: 27412066](#) — Ssa1 delivers precursors to mitochondrial import receptors; cytosolic factor at organellar surface.
- [PMID: 17636264](#) — Cytosolic Ssa1p modulates PDR5 transcription via nuclear Pdr3; supports cytosolic/nuclear pools.

Conclusion

The seed hypothesis is **partially supported**: its recommended action (KEEP_AS_NON_CORE) is a reasonable, conservative outcome, but its supporting rationale is wrong in a way that materially affects curation. There is no IBA supporting plasma membrane for Ssa1; the term rests entirely on one crude high-throughput proteomic fraction (HDA,

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and Ssa1 is an abundant cytosolic Hsp70 with no membrane-targeting sequence features. The curator should retain the term as non-core *or* remove it as a likely fractionation artifact — and, either way, correct the evidence code (HDA, not IBA) and the rationale text. The genuinely well-supported locations are cytosol, cytoplasm, and nucleus; any true surface pool is better captured by fungal-type cell wall (GO:0009277).

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